

Phase 3

Technical Architecture & Validation Fit

Date	27.06.2026
Team ID	NIL
Project Name	Software Installation Request Automation
Maximum Marks	4

Solution Architecture

Introduction

This phase details the technical realization of the project in ServiceNow Studio. It covers the creation of the scoped application, the Flow Designer implementation, and the Server-Side JavaScript Business Rule.

Problem–Solution Fit

The solution directly fits the problem by standardizing data intake (Catalog Item variables) and eliminating manual routing (Flow Designer).

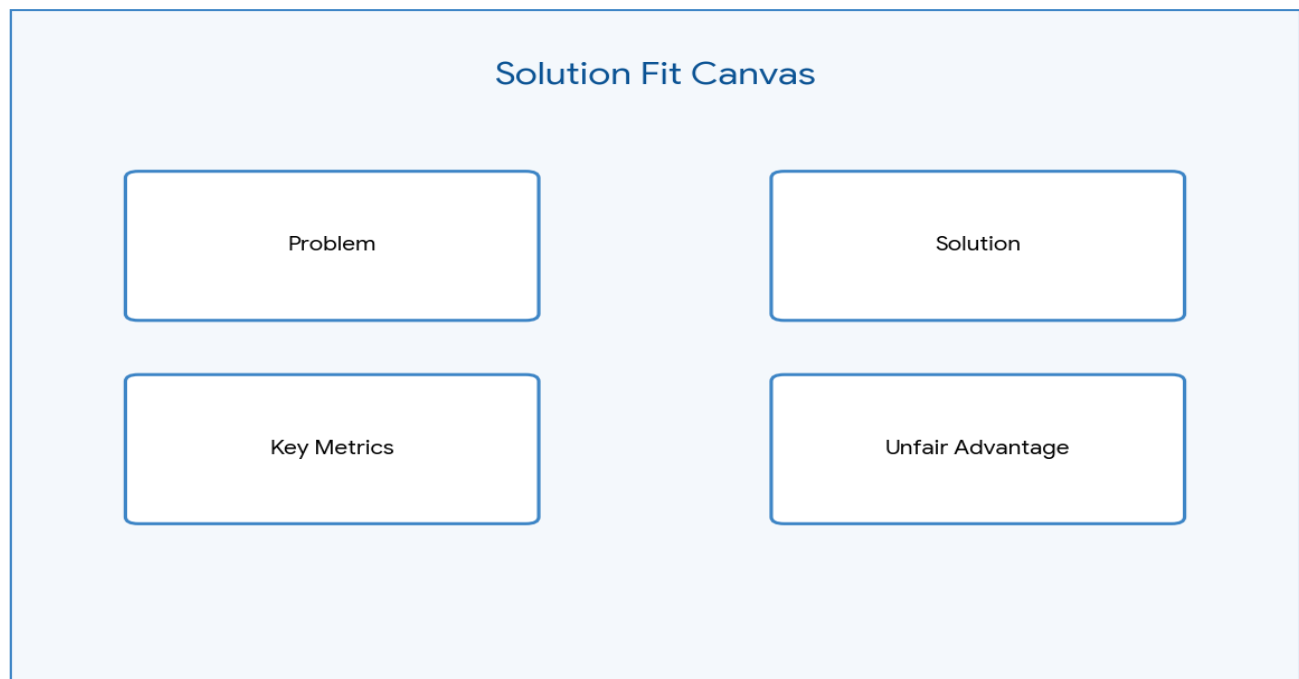


Figure: Solution Fit Canvas

Customer Segment

Internal corporate employees, line managers, and the IT 'Software Support' fulfillment team.

Jobs to be Done

1. Capture structured software requests. 2. Enforce managerial approval. 3. Dispatch installation tasks. 4. Monitor license availability.

Constraints

Must operate within a scoped application to prevent global namespace pollution. Must utilize out-of-the-box (OOTB) tables (REQ/RITM/SCTASK).

Proposed Solution

A ServiceNow Service Catalog item connected to a Flow Designer workflow, supplemented by an advanced Business Rule for exception handling.

Novelty

Pivoting from Legacy Workflow Editor to the modern Flow Designer ensures long-term platform support and easier maintainability.

Social Impact

Reduces IT friction, lowering workplace stress and improving employee satisfaction.

Business Model

Internal IT Service Management optimization; reduces operational expenditure (OpEx) by saving IT helpdesk hours.

Technology Stack

ServiceNow Platform, Flow Designer, Server-Side JavaScript (ES5 - Rhino engine), GlideRecord API.

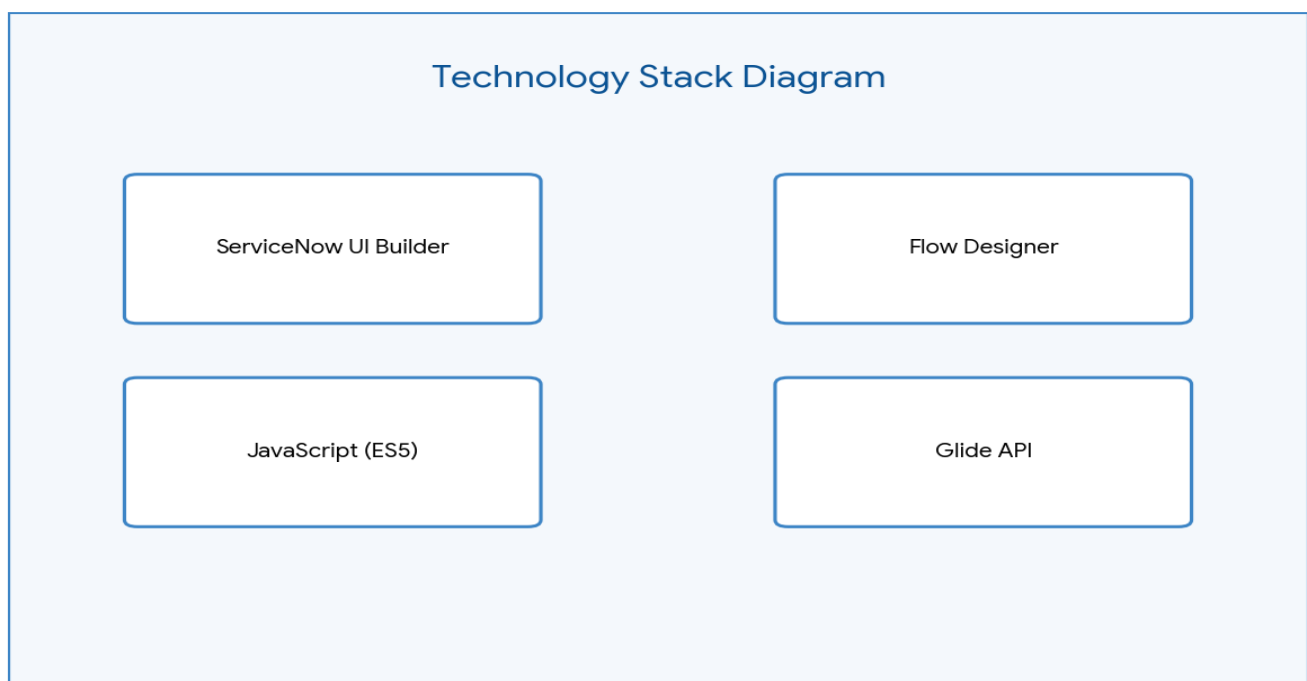


Figure: Technology Stack Diagram

Technical Architecture

Built entirely within the `Software Installation Request Automation` scoped app. Utilizes the Request Management architecture (sc_request -> sc_req_item -> sc_task).

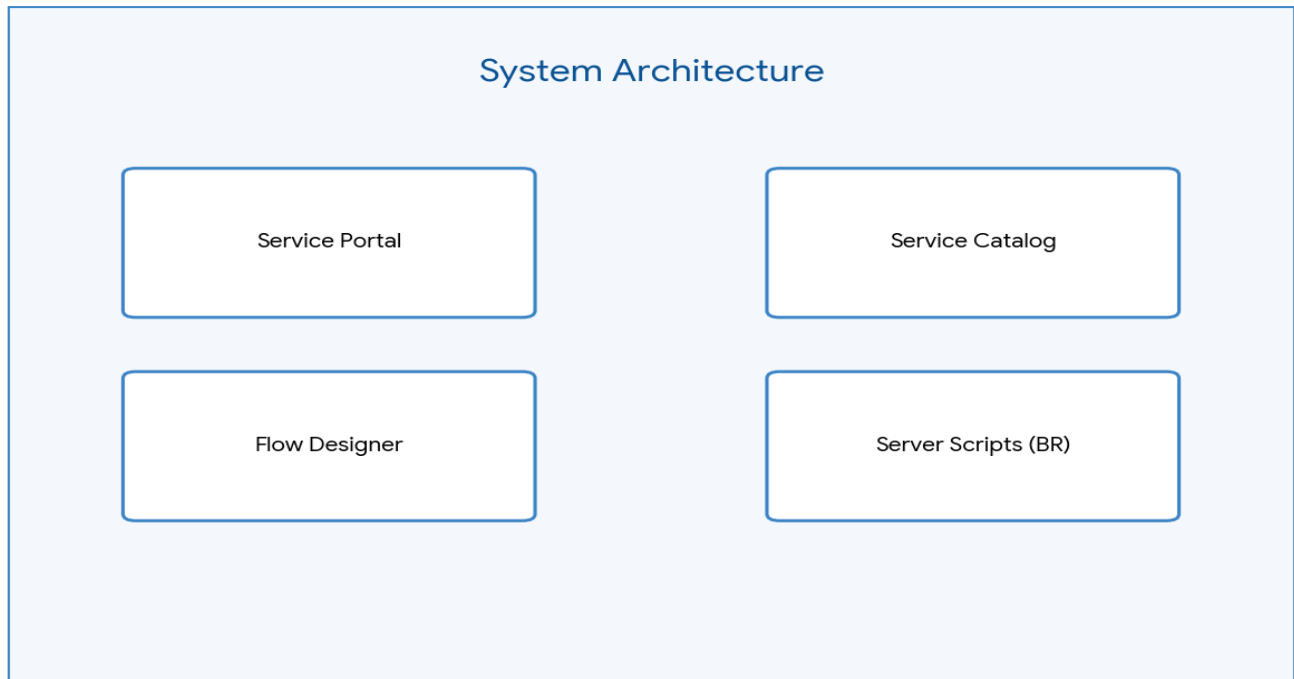


Figure: System Architecture

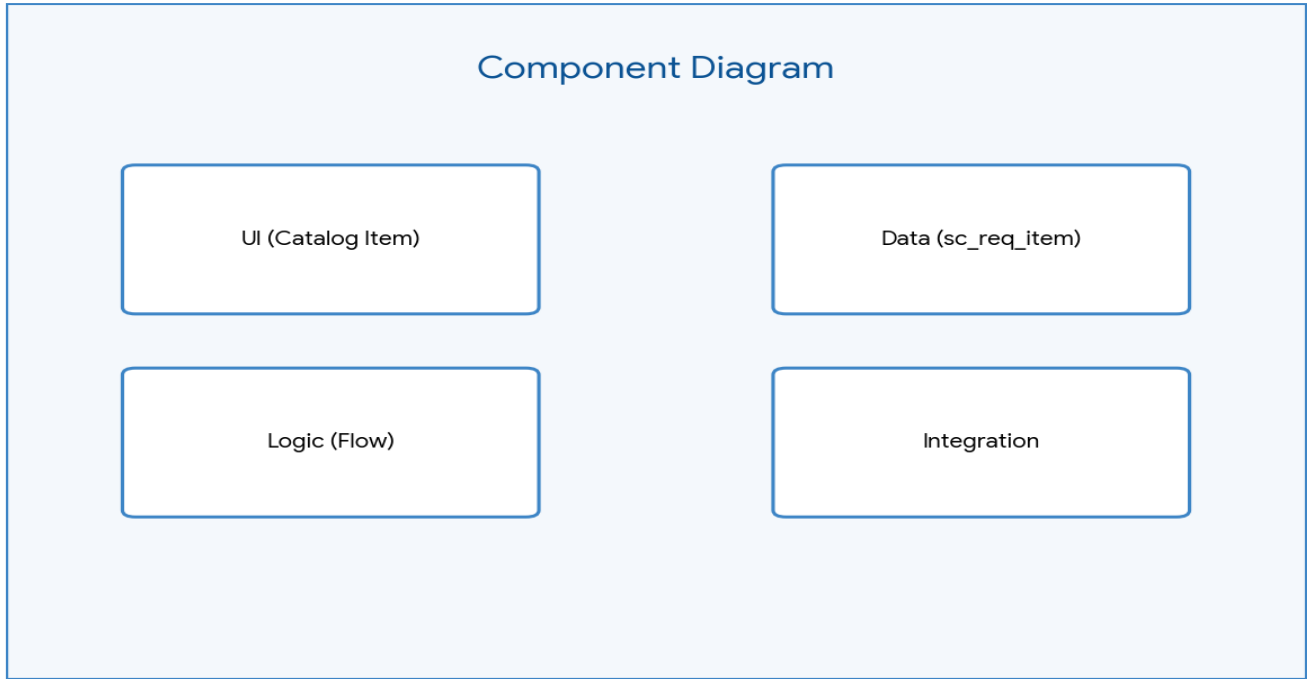


Figure: Component Diagram

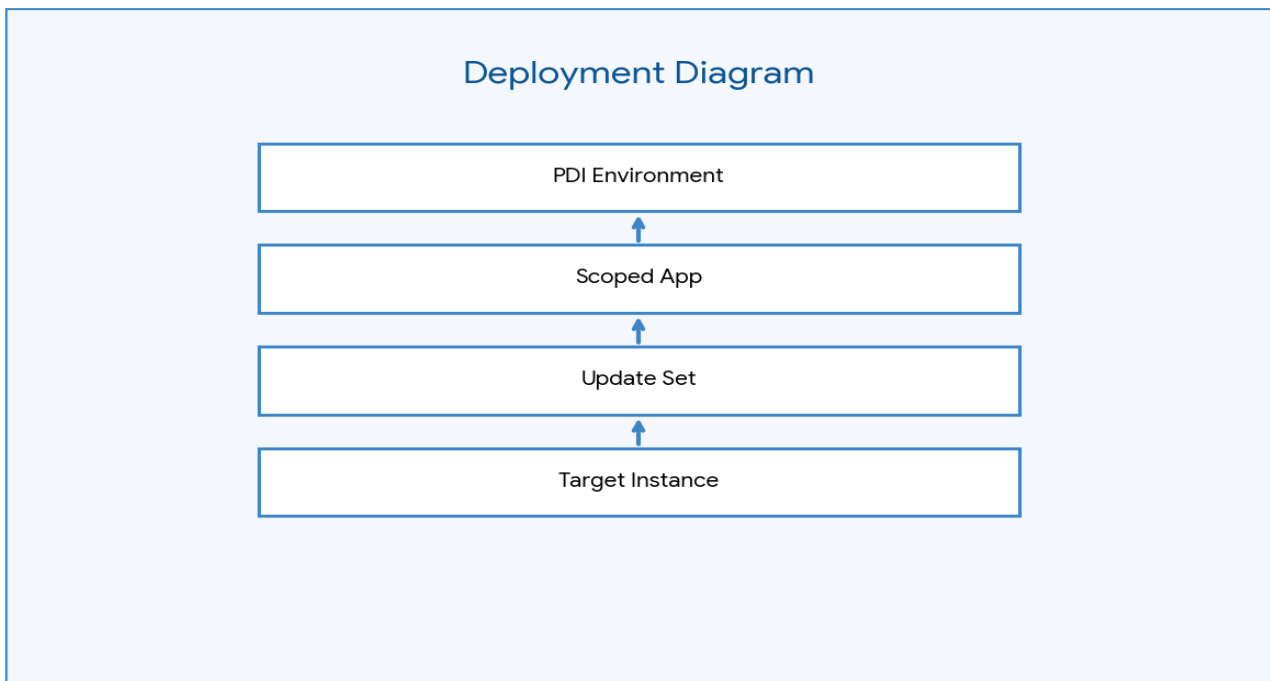


Figure: Deployment Diagram

Flow Designer Architecture

Trigger: Service Catalog. Action 1: Get Catalog Variables. Action 2: Ask for Approval (Software Support). Logic: If Approved -> Create Catalog Task (Wait for completion). Else -> Update Record (State = Closed Incomplete).

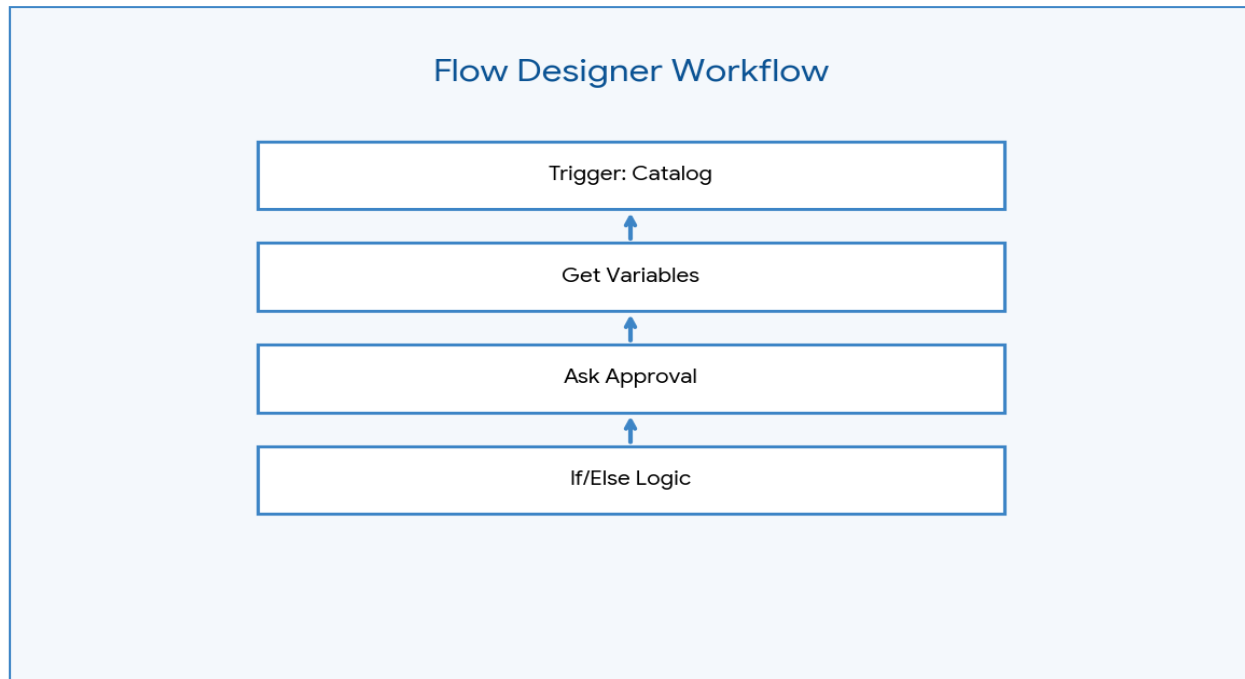


Figure: Flow Designer Workflow

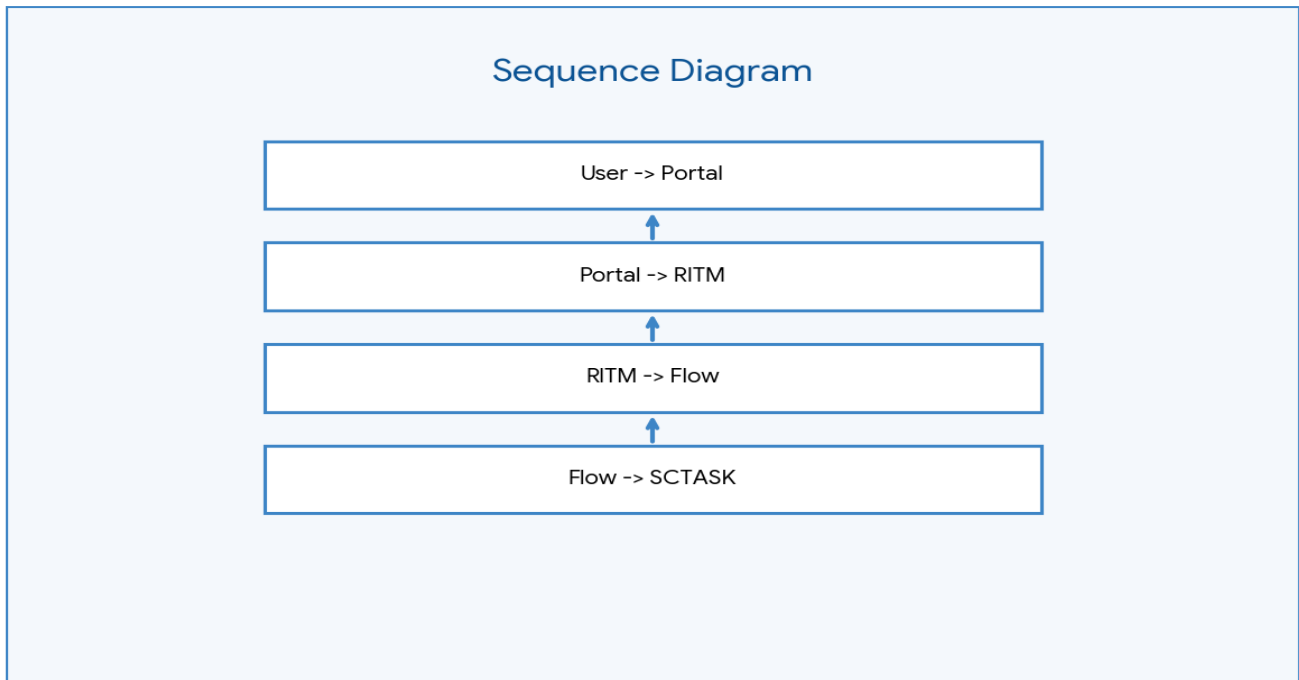


Figure: Sequence Diagram

Business Rule Logic

Advanced Business Rule on `sc\_req\_item`. When to run: After Update, Condition: State changes to 'Pending'. Script executes a GlideRecord insert on the `incident` table to report a license shortage.

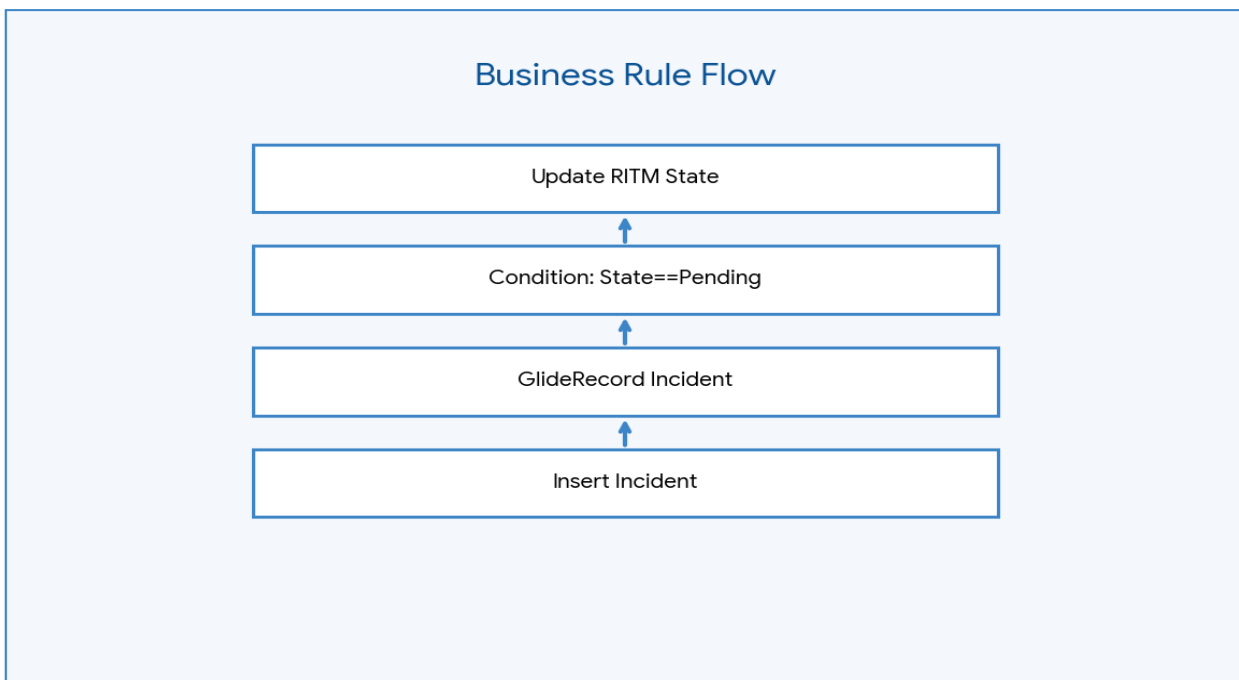


Figure: Business Rule Flow

Data Flow

Variables flow from the Portal to the RITM. Flow Designer extracts these variables and maps them to the generated SCTASK for the IT agent.

Security

Roles restrict who can approve tasks. The scoped application encapsulates logic securely. Variables are sanitized by the ServiceNow platform.

Scalability

Using Flow Designer allows seamless scaling. Additional approvals (e.g., Director level for high-cost software) can be added as sequential steps without coding.

Performance

Using an 'After' Business Rule ensures the database transaction completes quickly before the incident logic runs, preserving user-facing UI performance.

Phase Summary

Phase 3 successfully defined the core technical engine. The system leverages modern Flow Designer for standard operations and robust JavaScript for edge-case license issues.

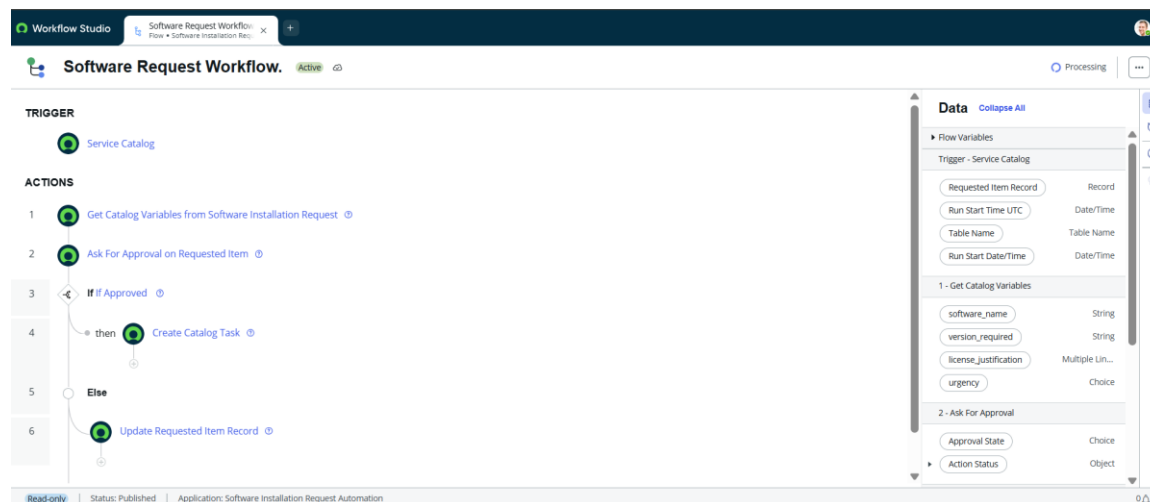


Figure: Flow Designer

Requested Item - RITM0010030

Number: RITM0010030
Item: Software Installation Request
Request: REQ0010028
Requested for: System Administrator
Due date: 2026-06-27 07:09:42
Configuration item:
Watch list:

Opened: 2026-06-27 07:09:42
Opened by: System Administrator
State: Request Approved
State: Open
Quantity: 1
Estimated delivery:
Backordered:
Order Guide:

Variables

What software do you need?
cursor
Specify version (if required)
2.0
Why do you need this software?
programming
Select urgency level
Normal

Comments (Customer visible)
Comments (Customer visible)
Work notes
Post

Activities: 1
System Administrator
Impact: 3 - Low
Opened by: System Administrator
State: Open
Field changes • 2026-06-27 07:09:42

Update Delete
Related Links
Flow Context
Catalog Tasks (1) Approvers (1)
State Search
Approval for • RITM0010030
State
Approved
Approver: Dharmisha Nallappan
Comments
Created: 2026-06-27 07:09:44

Figure: Approval Action

servicenow All Favorites History Workspaces Admin Catalog Task - SCTASK0010036 Search

Catalog Task SCTASK0010036

Number: SCTASK0010036
Assigned to:
Configuration item:
Active:

Approval: Not Yet Requested
Priority: -- None --
State: Open
Request item: RITM0010030
Requested for: System Administrator

Variables

What software do you need?
cursor
Specify version (if required)
2.0
Why do you need this software?
programming
Select urgency level
Normal

Short description: Fulfill Software Request
Description:

Figure: Catalog Task

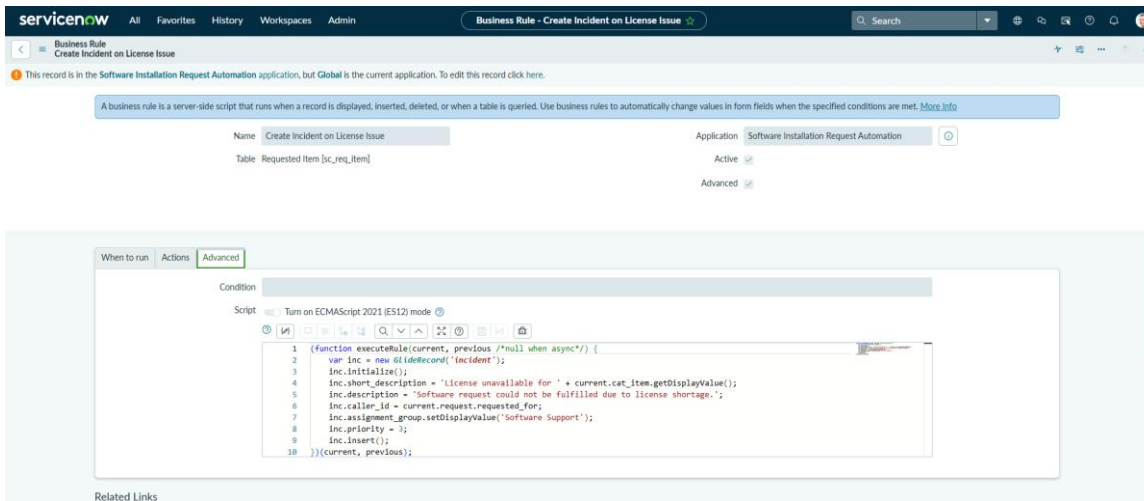


Figure: Business Rule

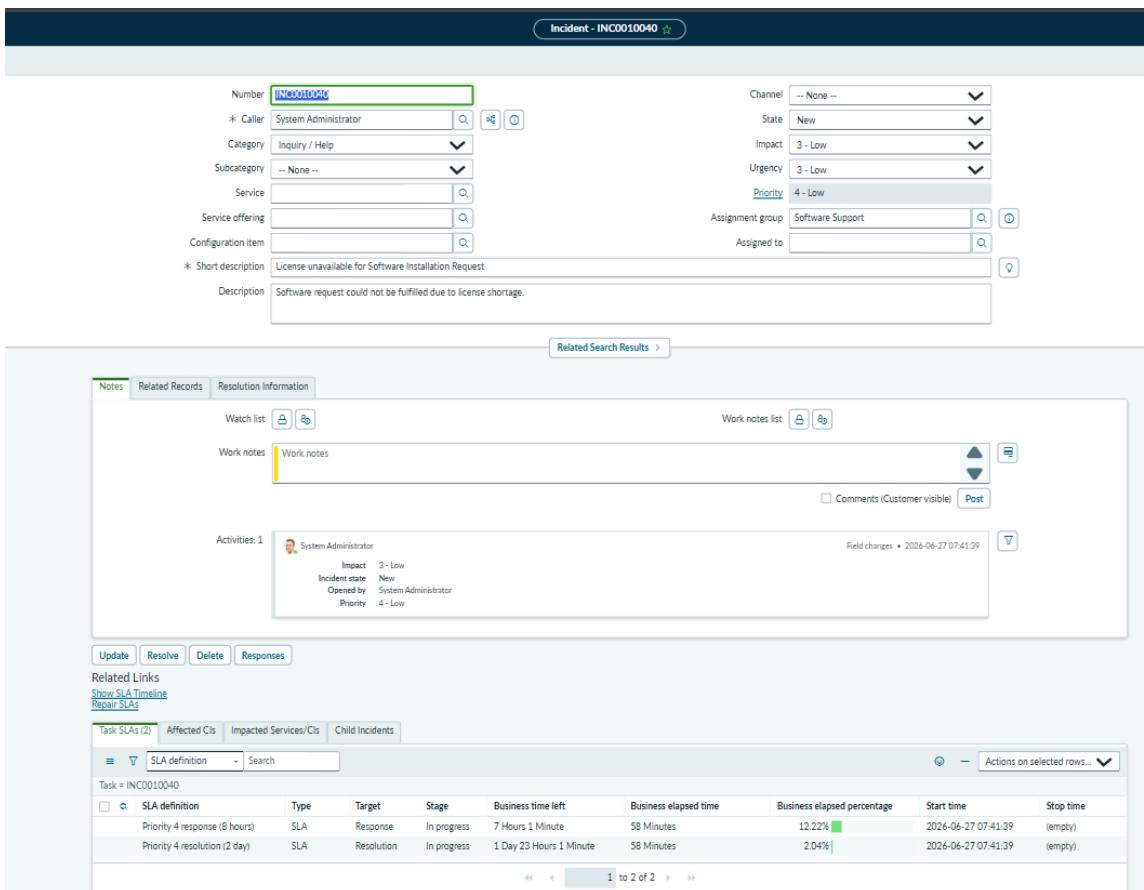


Figure: Incident Creation